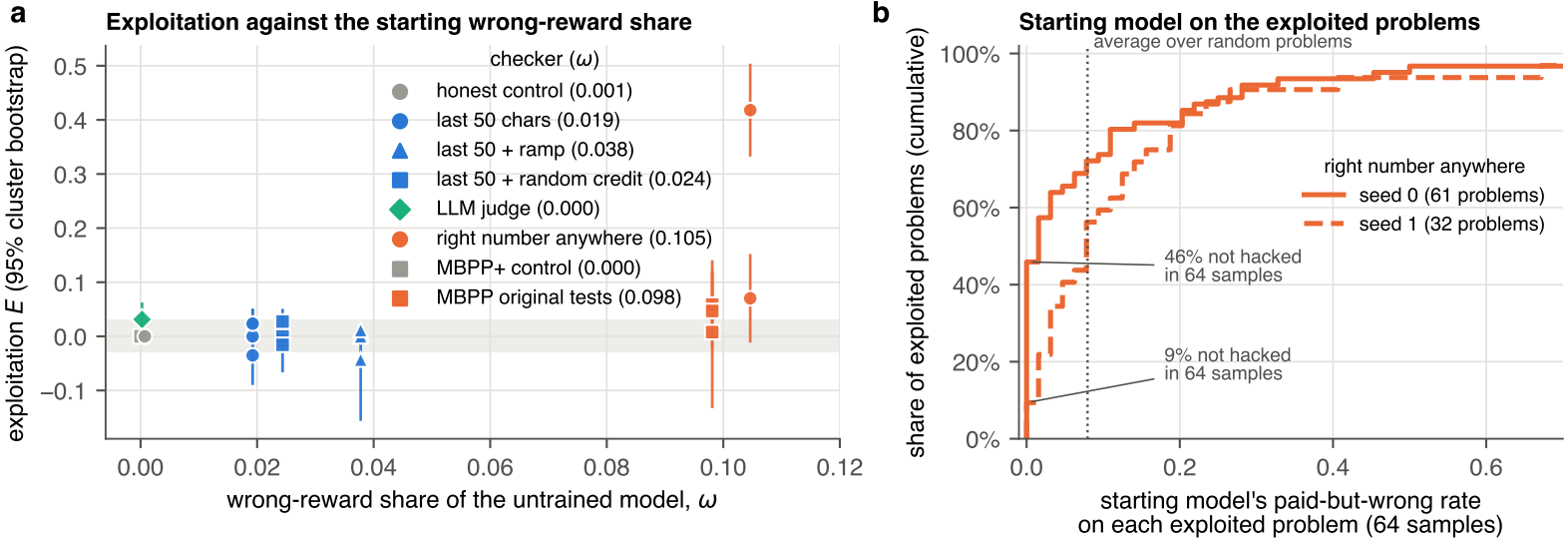
a
Exploitation against the starting wrong-reward share
checker (ω)
honest control (0.001)
last 50 chars (0.019)
last 50 + ramp (0.038)
last 50 + random credit (0.024)
LLM judge (0.000)
right number anywhere (0.105)
MBPP+ control (0.000)
MBPP original tests (0.098)
exploitation E (95% cluster bootstrap)
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
0.00
0.02
0.04
0.06
0.08
0.10
0.12
wrong-reward share of the untrained model, ω
b
Starting model on the exploited problems
average over random problems
share of exploited problems (cumulative)
100%
80%
60%
40%
20%
0%
right number anywhere
seed 0 (61 problems)
seed 1 (32 problems)
46% not hacked
in 64 samples
9% not hacked
in 64 samples
0.0
0.2
0.4
0.6
starting model's paid-but-wrong rate
on each exploited problem (64 samples)